

Write Equations

Flagship

Lesson 7-1-flagship

Name: _____

Date: _____

Class: _____

COLD CASE MISSION

| The Equation Vault

You are Detective Ruiz's new code analyst. A tip just came in: 'A number of stolen gems plus 8 more equals 20 total.' The evidence vault only opens when the clue is translated into a correct equation. Master writing equations and the case cracks wide open.

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$$x + 2 = 7$$

balanced with =

$x + 5 = 12$ means 'some number plus 5 equals 12'

Equation

Write the definition:

x

stands for a number

In $n + 3 = 10$, the letter n stands for the unknown number ($n = 7$)

Variable

Write the definition:

$$x + 2 = 7$$

balanced with =

$7 + 3 = 10$ — both sides equal 10

Equal sign

Write the definition:

$$3x + 5$$

no equal sign

$2x + 5$ is an expression; it becomes an equation when you write $2x + 5 = 15$

Expression

Write the definition:

Guided Notes Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can write an equation to represent a real-world situation.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Equation

Variable

Equal sign

Expression

1 A math sentence that says two amounts are equal, with an = sign, is an _____.

2 A letter that stands for an unknown number is a _____.

3 The symbol = that shows two amounts are the same is the _____.

4 A group of numbers and variables with no equal sign is an _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.



I do — watch

Follow each step as your teacher solves it.

Problem: Which equation represents: 'A number plus 12 equals 30'?

A. $n + 12 = 30$

B. $n - 12 = 30$

C. $12n = 30$

D. $n / 12 = 30$

Step 1

'Plus' means addition, so the equation is $n + 12 = 30$.



Answer: A. $n + 12 = 30$

 We do — together

Solve this one with your class using the same steps.

Problem: Which equation represents: 'Seven less than a number is 18'?

A. $n - 7 = 18$

B. $7 - n = 18$

C. $n + 7 = 18$

D. $n / 7 = 18$

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: Which equation represents: 'Three times a number equals 21'?

A. $3n = 21$

B. $n + 3 = 21$

C. $n - 3 = 21$

D. $n / 3 = 21$

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Ledger note: 'Six times the number of stolen coins equals 54.' Which equation models this clue?

- A. $6c = 54$
- B. $c + 6 = 54$
- C. $c - 6 = 54$
- D. $c / 6 = 54$

Show your work:

2. Vault log: 'A number of gold bars shared equally into 5 cases puts 4 bars in each case.' Which equation models this clue?

- A. $b / 5 = 4$
- B. $5b = 4$
- C. $b - 5 = 4$
- D. $b + 5 = 4$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Write a real-world situation that can be modeled by the equation $n / 5 = 9$. Explain what n represents and verify your equation makes sense.

Sentence starter: Situation: _____. In this problem, n represents _____. Check: $n = \underline{\hspace{1cm}}$ because $\underline{\hspace{1cm}} \div 5 = 9$.

Show your work:

Reflect — Exit Ticket

Which equation represents: 'A number divided by 6 equals 9'?

- A. $n / 6 = 9$
- B. $6n = 9$
- C. $n - 6 = 9$
- D. $n + 6 = 9$

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Equation
2. **Notes 2:** Variable
3. **Notes 3:** Equal sign
4. **Notes 4:** Expression
5. **We Try (notes):** A. $n - 7 = 18$ — *'Seven less than a number' means start with the number and subtract 7: $n - 7 = 18$.*
6. **You Try (notes):** A. $3n = 21$ — *'Three times' means multiply by 3, so the equation is $3n = 21$.*
7. **Try It 1:** A. $6c = 54$ — *Let c be the number of stolen coins. 'Six times' means multiply by 6, so the equation is $6c = 54$.*
8. **Try It 2:** A. $b / 5 = 4$ — *Let b be the total gold bars. Sharing equally into 5 cases means $b / 5$, and each case has 4, so $b / 5 = 4$.*
9. **Exit Ticket:** A. $n / 6 = 9$ — *'Divided by 6' means division: $n / 6 = 9$. Check: $54 / 6 = 9$ ✓*